Dataquest Challenge 2024: Personalised Solutions

Data Dictionary

This document shows a brief overview and explanation of the dataset that is used for the Dataquest challenge. A screenshot of the top set of rows are shown in Figure 1.

idcol	interaction	int_date	item	page	tod	item_type	item_descrip	segment	beh_segment	active_ind
755	DISPLAY	17-Jan-23	NONE	Screen1	Afternoon	ALL		segment3	B01	Semi Active
4521	DISPLAY	27-Feb-23	NONE	Screen1	Afternoon	ALL		segment1	B07	Semi Active
4521	DISPLAY	18-Feb-23	NONE	Screen1	Afternoon	ALL		segment1	B07	Semi Active
4521	DISPLAY	30-Jan-23	NONE	Screen1	Morning	ALL		segment1	B07	Semi Active
4521	CLICK	5-Feb-23	IBAB	Screen1	Afternoon	INSURE	GENERIC MESSAGE	segment1	B07	Semi Active
4521	CHECKOUT	5-Feb-23	IBAB	Screen1	Afternoon	INSURE	GENERIC MESSAGE	segment1	B07	Semi Active
6145	DISPLAY	26-Feb-23	NONE	Screen1	Evening	ALL		segment3	B01	Cold Start
6145	DISPLAY	27-Jan-23	NONE	Screen1	Early	ALL		segment3	B01	Cold Start
6145	DISPLAY	10-Feb-23	NONE	Screen1	Morning	ALL		segment3	B01	Cold Start
6145	DISPLAY	10-Jan-23	NONE	Screen1	Afternoon	ALL		segment3	B01	Cold Start
7125	DISPLAY	18-Feb-23	NONE	Screen1	Morning	ALL		segment3	B01	Cold Start
8469	DISPLAY	5-Feb-23	NONE	Screen1	Morning	ALL		segment1	B01	Semi Active
8469	DISPLAY	14-Jan-23	NONE	Screen1	Afternoon	ALL		segment1	B01	Semi Active
8469	DISPLAY	21-Feb-23	NONE	Screen1	Morning	ALL		segment1	B01	Semi Active
13768	DISPLAY	2-Feb-23	NONE	Screen1	Morning	ALL		segment3	B01	Cold Start
14454	DISPLAY	9-Feb-23	NONE	Screen1	Morning	ALL		segment2	B01	Active
14454	CLICK	8-Feb-23	CAFM	Screen2	Morning	TRANSACT	FUSION MIGRATION	segment2	B01	Active
14454	CHECKOUT	8-Feb-23	CAFM	Screen2	Morning	TRANSACT	FUSION MIGRATION	segment2	B01	Active

Figure 1 Screenshot of the top rows of the dq_ps_challenge_v2.csv dataset.

Each of the features columns are grouped into these types of feature-types, to fit into the user-item interaction matrix as in ***Figure 2***.

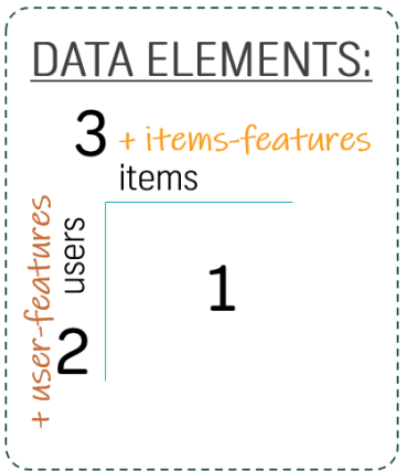

Figure 2 The user-item-interaction matrix, and related features.

The feature type and description of each column in the dataset, are detailed in and it's contents is depicted in Table 1.

Table 1 The Data Dictionary for the Personalised Solutions Dataquest challenge.

Column	Feature-type	Description
Idcol	User-identifier (2)	The id-column that is a unique identifier per customer
interaction	Interaction (1)	Three interactions: **DISPLAY**: this refers to a customer logging in, but no specific click or checkout of a item **CLICK**: a click-interaction, showing that a customer have clicked on one of the items that were displayed. This shows some initial interest in a product, but is not the final CHECKOUT interaction yet. **CHECKOUT**: this is a final step to take-up a product that were displayed to a customer.
int_date	Interaction-date (1)	The date on which the interaction occurred
item	Item-identifier (3)	The item-code with which the customer interacted with. The items are grouped into the *item_type* column, and more details of what the code refers to is in the *item_description* column. When there was only a DISPLAY interaction (i.e. no CLICK or CHECKOUT interaction), then the item is the **NONE** class.
page	context	There are two pages on the App via which the customer can interact
tod	context	Time-of-day: The time of each interaction was grouped into 4 timeslots: **Early**: 00:00 -<06:00 **Morning**: 06:00-<12:00 **Afternoon**: 12:00-<18:00 **Evening**: 18:00-<00:00
item_type	Item-feature (3)	Item_type: All of the items are grouped into the following groups: **TRANSACT, LEND, INVEST, INSURE, CONNECT** and **LIFESTYLE** When the interaction is DISPLAY, the item_type reflects as ALL
item_descrip	Item-feature (3)	A description of each item
segment	user-feature (2)	4 broad customer segments, broadly linked to income, with segment1 to segment 4 ranging from lower to higher income
beh_segment	user-feature (2)	50 detailed customers segments. The exact definition is not disclosed, but can be seen as 50 different customer segments.
active_ind	user-feature (2)	**Active**: customers who interact regularly on this platform **Semi Active**: customers who interact occasionally on this platform **Cold Start**: customers who interact very rarely on this platform, or not at all. *This definition was applied as at the start of the data time-window (1 January 2023).*